

ABASYN UNIVERSITY PESHAWAR

DEPARTMENT OF COMPUTER

Assignment # (6)

Subject :-	**SOFTWARE TESTING**
Class / section :-	**CB**
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1- Program Pseudocode:

```
INPUT membership
INPUT purchaseAmount
INPUT coupon

A = (membership == "Premium")
B = (purchaseAmount >= 10000)
C = (coupon == TRUE)

IF (A AND (B OR C)) THEN
    PRINT "Discount Applicable"
ELSE
    PRINT "No Discount"
END IF
```

Part 1: Decision Coverage (Branch Coverage)

1. Identification of the Decision

Definition: In structural white-box testing, a decision (or branch predicate) is a Boolean expression that determines which control-flow path executes.

In this pseudocode, the decision is the entire expression evaluated by the IF statement:

Decision (D) = A AND (B OR C)

This decision produces two mutually exclusive outcomes:

- **Decision = TRUE (True Branch):** A is TRUE and either B or C is TRUE. The program prints "Discount Applicable".
- **Decision = FALSE (False Branch):** A is FALSE, or both B and C are FALSE. The program prints "No Discount".

2. Test Case Table for Decision Coverage

Decision coverage requires the overall decision to take both possible outcomes at least once. These two cases exercise the TRUE and FALSE branches.

Test Case ID	Membership	Purchase Amount	Coupon Status	Decision Result	Expected Output	Branch Covered
TC_DC_01	Premium	10000	FALSE	**TRUE**	Discount Applicable	TRUE branch
TC_DC_02	Regular	12000	TRUE	**FALSE**	No Discount	FALSE branch

Measurement & Verification: Decision Coverage = (decision outcomes tested / total decision outcomes) x 100% = (2 / 2) x 100% = **100%**. Both TRUE and FALSE branches are exercised.

Part 2: Condition Coverage (Branch Condition Coverage)

1. Identification of Individual Conditions

Definition: A condition is an atomic Boolean predicate that evaluates independently to TRUE or FALSE. The decision contains three such conditions:

- **Condition A:** membership == "Premium". TRUE for Premium membership; FALSE otherwise.
- **Condition B:** purchaseAmount >= Rs. 10,000. TRUE at or above Rs. 10,000; FALSE below Rs. 10,000.
- **Condition C:** coupon == TRUE. TRUE when a valid promotional coupon is present; FALSE otherwise.

2. Test Case Table for Condition Coverage

Condition coverage requires every individual condition to evaluate to TRUE and FALSE at least once. The following three test cases provide both outcomes for A, B, and C.

Test Case ID	Membership	Amount (Rs.)	Coupon	A	B	C	Decision D	Expected Output
TC_CC_01	Premium	12000	TRUE	**TRUE**	**TRUE**	**TRUE**	**TRUE**	Discount Applicable
TC_CC_02	Premium	8000	FALSE	**TRUE**	**FALSE**	**FALSE**	**FALSE**	No Discount
TC_CC_03	Regular	12000	TRUE	**FALSE**	**TRUE**	**TRUE**	**FALSE**	No Discount

Coverage check: A is TRUE in TC_CC_01/02 and FALSE in TC_CC_03. B is TRUE in TC_CC_01/03 and FALSE in TC_CC_02. C is TRUE in TC_CC_01/03 and FALSE in TC_CC_02. Therefore all six required condition outcomes are covered: (6 / 6) x 100% = **100%**.

Part 3: Decision/Condition Coverage (D/C Coverage)

1. Designing a Test Suite Achieving Both Decision and Condition Coverage

Definition: Decision/Condition Coverage (D/C Coverage) combines the two criteria. The test set must exercise both overall decision outcomes and both outcomes of each individual condition.

- **Decision requirement:** D must evaluate to TRUE and FALSE at least once.
- **Condition requirement:** A, B, and C must each evaluate to TRUE and FALSE at least once.

The three-case suite below covers the decision and every individual condition:

Test Case ID	Membership	Amount (Rs.)	Coupon	A	B	C	Decision D	Expected Output	Branch
TC_DCC_01	Premium	12000	TRUE	**TRUE**	**TRUE**	**TRUE**	**TRUE**	Discount Applicable	THEN
TC_DCC_02	Premium	8000	FALSE	**TRUE**	**FALSE**	**FALSE**	**FALSE**	No Discount	ELSE
TC_DCC_03	Regular	12000	TRUE	**FALSE**	**TRUE**	**TRUE**	**FALSE**	No Discount	ELSE

2. Traceability: How Each Test Case Contributes to Required Coverage

Test Case	Condition A contribution	Condition B contribution	Condition C contribution	Decision contribution	Cumulative coverage
TC_DCC_01	A = TRUE	B = TRUE	C = TRUE	D = TRUE	A/B/C: 1 of 2; D: 1 of 2
TC_DCC_02	A = TRUE	B = FALSE	C = FALSE	D = FALSE	A 1/2; B,C,D 2/2
TC_DCC_03	A = FALSE	B = TRUE	C = TRUE	D = FALSE	A/B/C/D: 2 of 2

- **Decision verification:** TRUE is covered by TC_DCC_01; FALSE is covered by TC_DCC_02 and TC_DCC_03. Decision coverage = 2/2 = 100%.
- **Condition verification:** A, B, and C each take both TRUE and FALSE across the three cases. Condition coverage = 6/6 = 100%.

- **Combined verification:** The same three cases satisfy both criteria, so Decision/Condition Coverage is complete at 100%.

Part 4: Multiple Condition Coverage (MCC)

1. All Possible Condition Combinations

There are three independent Boolean conditions. The total number of combinations is 2^3 = **8**. Multiple Condition Coverage tests every combination exactly once.

2. Complete Combination Table

Case #	A: Premium	B: Amount >= 10000	C: Coupon valid	Decision A AND (B OR C)	Expected Output	Result
1	TRUE	TRUE	TRUE	**TRUE**	Discount Applicable	Discount
2	TRUE	TRUE	FALSE	**TRUE**	Discount Applicable	Discount
3	TRUE	FALSE	TRUE	**TRUE**	Discount Applicable	Discount
4	TRUE	FALSE	FALSE	**FALSE**	No Discount	No discount
5	FALSE	TRUE	TRUE	**FALSE**	No Discount	No discount
6	FALSE	TRUE	FALSE	**FALSE**	No Discount	No discount
7	FALSE	FALSE	TRUE	**FALSE**	No Discount	No discount
8	FALSE	FALSE	FALSE	**FALSE**	No Discount	No discount

Interpretation: Discount is applicable only when A is TRUE and at least one of B or C is TRUE. Thus combinations 1, 2, and 3 produce "Discount Applicable"; combinations 4 through 8 produce "No Discount".

Part 5: Coverage Calculation and Comparison

Technique	Coverage calculation	Coverage
Decision Coverage	(2 decision outcomes tested / 2 total) x 100%	100%
Condition Coverage	(6 condition outcomes tested / 6 total) x 100%	100%
Decision/Condition Coverage	Both decision outcomes and all six condition outcomes	100%
Multiple Condition Coverage	(8 combinations tested / 8 possible) x 100%	100%

Coverage technique comparison:

- **Decision coverage** checks that the whole IF expression takes each outcome (TRUE and FALSE).
- **Condition coverage** checks that each atomic condition A, B, and C independently takes both TRUE and FALSE values.
- **Decision/Condition coverage** requires both decision outcomes and both outcomes of every individual condition in one test suite.
- **Multiple Condition Coverage** tests every possible combination of individual condition values; for three conditions it requires eight cases.

MCC subsumes individual condition outcome coverage and decision coverage for this decision, but requires more tests. Decision/Condition coverage uses fewer tests than MCC while explicitly checking the two stated criteria.